Cross-Lingual Natural Language Inference with XLM-RoBERTa

Final Project Report for the course "Introduction to Computational Linguistics"

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1. Introduction and Task Description

For this project, I selected the Option B: Two Languages + One Model configuration. The objective was to fine-tune a multilingual pre-trained language model on a downstream task using data from both English and Turkish to analyze its cross-lingual capabilities.

The Downstream Task: Natural Language Inference involves determining the logical relationship between a premise and a hypothesis. The model classifies this relationship into one of three labels:

- Entailment (0): The hypothesis logically follows from the premise.
- Neutral (1): The truth of the hypothesis cannot be determined from the premise.
- Contradiction (2): The hypothesis contradicts the premise.

Dataset: I utilized the XNLI (Cross-lingual Natural Language Inference) dataset. Turkish was selected as the second language to evaluate the model's performance on an agglutinative language with distinct morphological characteristics compared to English.

Setup: Google Colab, utilizing an L4 GPU.

2. Adaptations in the Code

I followed all orientations from Tutorials 3 and 4. But I implemented some adaptations to suit the nature of the task.

1. Mixed-Language Validation: I constructed a custom validation set by concatenating English and Turkish examples using the datasets library. This allowed me to monitor the model's generalization capabilities across *both* languages simultaneously during training.
2. Real-time Logging: I enabled report_to="tensorboard" to generate real-time loss curves, allowing for the immediate diagnosis of overfitting.
3. Reproducibility (Seeding): I enforced strict reproducibility by manually setting random seeds across Python, NumPy, and PyTorch, ensuring that the improvements observed in Experiment 4 were due to data size and not random weight initialization.

4. Embedding Extraction (RoBERTa specific): I adapted the hidden-state extraction logic to index position [0] directly. The course tutorial looked up the [CLS] token ID from BERT's vocabulary, which does not exist in XLM-RoBERTa's SentencePiece tokenizer — its equivalent start token is <s>, always at position 0 after tokenization.

3. Experimental Process

The four experiments reported here were selected from a broader exploration to isolate the most instructive contrasts: baseline vs. regularization, baseline vs. learning rate/epochs, and baseline vs. training budget.

Experiment 1 : Standard fine-tuning parameters. Resulted in moderate accuracy (~64%) but signs of overfitting.

Experiment 2: Higher weight decay and dropout to combat overfitting, but it resulted in lower accuracy (~60.5%) and higher loss.

Experiment 3: I increased training to 6 epochs with a lower learning rate. Clear overfitting emerged: while training loss decreased steadily, validation loss spiked significantly after epoch

Experiment 4 (Best Model): I increased the training data size to 10,000 examples (balanced between languages). This comparison isolated data size as a variable while holding the other hyperparameters constant relative to Experiment 1, confirming that the primary bottleneck was data, not hyperparameter choice.

4. Quantitative Results

The table below summarizes the performance of the model on the mixed validation set:

Experiment	Validation Accuracy	Validation Loss	Observation
Exp 1	64.18%	0.8916	Overfitting
Exp 2	60.50%	0.9710	Underfitting / Too restricted

Exp 3	63.51%	0.9447	Severe Overfitting
Exp 4	69.92%	0.8803	Best Performance

Final Evaluation on Test Set

After selecting the model from Experiment 4, I performed the final evaluation on the official XNLI Test Set (data never seen during training).

Language	**Test Set Accuracy**
English (Source)	72.36%
Turkish (Target)	66.21%

There is a performance gap of approximately 6.15% between English and Turkish. I believe the accuracy in Turkish shows a good transfer of semantic logic, even when the model was pre-trained and fine-tuned with a bias toward English linguistic structures.

5. Qualitative Results and Linguistic Analysis

A recurring pattern across errors was a systematic bias toward predicting Contradiction, especially in cases where the semantic relationship between premise and hypothesis required discourse-level inference or morphological decomposition. When strong lexical overlap was absent, the model appeared to default to contradiction rather than neutral or entailment.

One example concerns the effect of agglutination and discourse interpretation:

Premise: *"Ve Anne, evdeyim dedi."*
Hypothesis: *"Okul servisi onu bırakır bırakmaz annesini aradı."*
Gold Label: Neutral | Prediction: Contradiction

In the hypothesis, the noun *annesini* consists of the root *anne* (mother) combined with the possessive suffix *-(s)i* and the accusative suffix *-(n)i*, where *s* and *n* function as buffer consonants. This morphological complexity likely contributes to token fragmentation during subword tokenization, weakening lexical alignment between *Anne* in the premise and *annesini* in the hypothesis.

However, feedback from a native speaker reveals that the premise itself is pragmatically ambiguous. The sentence *"Ve Anne, evdeyim dedi"* can plausibly be interpreted as reported speech equivalent to *"And [they] said, 'Mom, I'm home.'"* Under this interpretation, *Anne* functions as an addressee rather than a third-person referent, implying that the speaker communicated with their mother. This reading establishes a clear semantic connection with the hypothesis, in which the subject explicitly calls their mother. As a result, the gold label Neutral becomes debatable, and an Entailment reading is at least plausible. The model's error, therefore, cannot be attributed solely to morphological mismatch.

A second set of errors involves modality and logical inference, exemplified by the following pair:

Premise: *"Gitmesine gerek yoktu."*
Hypothesis: *"Katılması yasaktı."*
Gold Label: Entailment | Prediction: Contradiction

At a lexical level, the premise uses *gitmek* ("to go"), while the hypothesis uses *katılmak* ("to join"), two morphologically unrelated verbs. More importantly, the premise expresses a lack of necessity (*gerek yoktu*), whereas the hypothesis expresses prohibition (*yasaktı*). While it is logically true that if an action is forbidden, it is unnecessary to perform it, this inference is not straightforward in natural language use.

Gitmek in Turkish is directionally ambiguous: it can mean either "to go to" a place or "to leave" a place, depending on context and case marking. For example, *"Toplantıya gitmesine gerek yoktu"* ("They didn't need to go to the meeting") combined with *"Katılması yasaktı"* ("It was forbidden for them to join") yields a relation closer to Neutral. In contrast, *"Toplantıdan gitmesine gerek yoktu"* ("They didn't need to leave the meeting") paired with *"Katılması yasaktı"* results in a genuine contradiction. Since the original premise lacks explicit directional marking, the gold label Entailment presupposes a specific interpretation that is not uniquely determined by the sentence itself.

Regarding negation, in Turkish, it is expressed morphologically via the suffix *-ma/-me*, as in *bilmiyordum* ("I did not know") and *gitmedim* ("I did not go"). In several cases, the presence of negation in both premise and hypothesis appeared to disproportionately attract the model's attention. The model often treated two negative statements as mutually incompatible, resulting in incorrect Contradiction predictions even when the true relation was Neutral.

It is also important to note a structural difference between the languages: English sentences in the dataset are typically longer in word count, while Turkish sentences are more concise and morphologically dense, which creates an alignment challenge for XLM-R. Because XNLI Turkish is derived from machine translation of the English original, the Turkish side also inherits cross-linguistic

asymmetries: gendered pronouns in English (*he*, *she*) collapse into Turkish's gender-neutral *o*, which can erase distinctions that the English gold label implicitly relied on. This is a known limitation of translation-based multilingual benchmarks rather than a fault of the model.

6. Visual Analysis

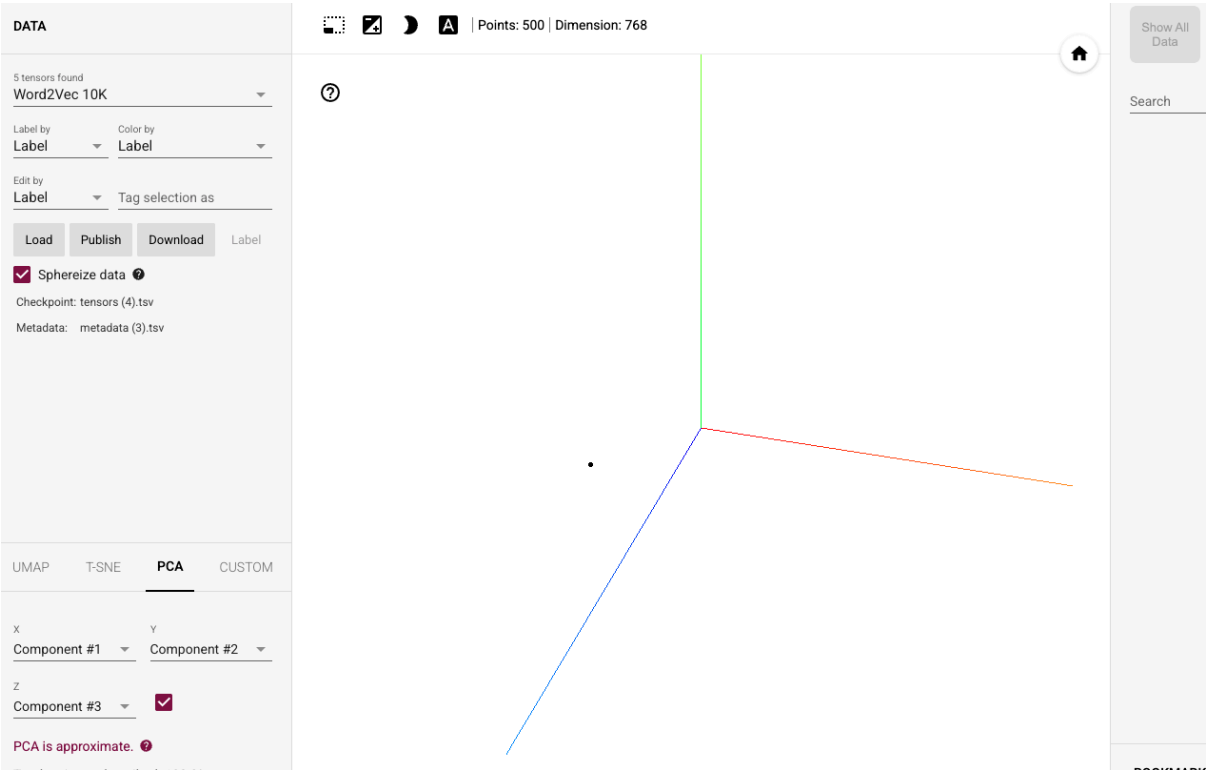

Figure 1: Layer 0 (Input Embeddings)

This visualization displays the PCA projection of the input embeddings (layer 0). The projections collapse into a dense region near the origin — not because the representations are identical, but because at the input stage the variance relevant to NLI is not yet encoded: differences between premise-hypothesis pairs exist in the token embeddings, but are not aligned with any task-specific axis that PCA can surface as a dominant component.

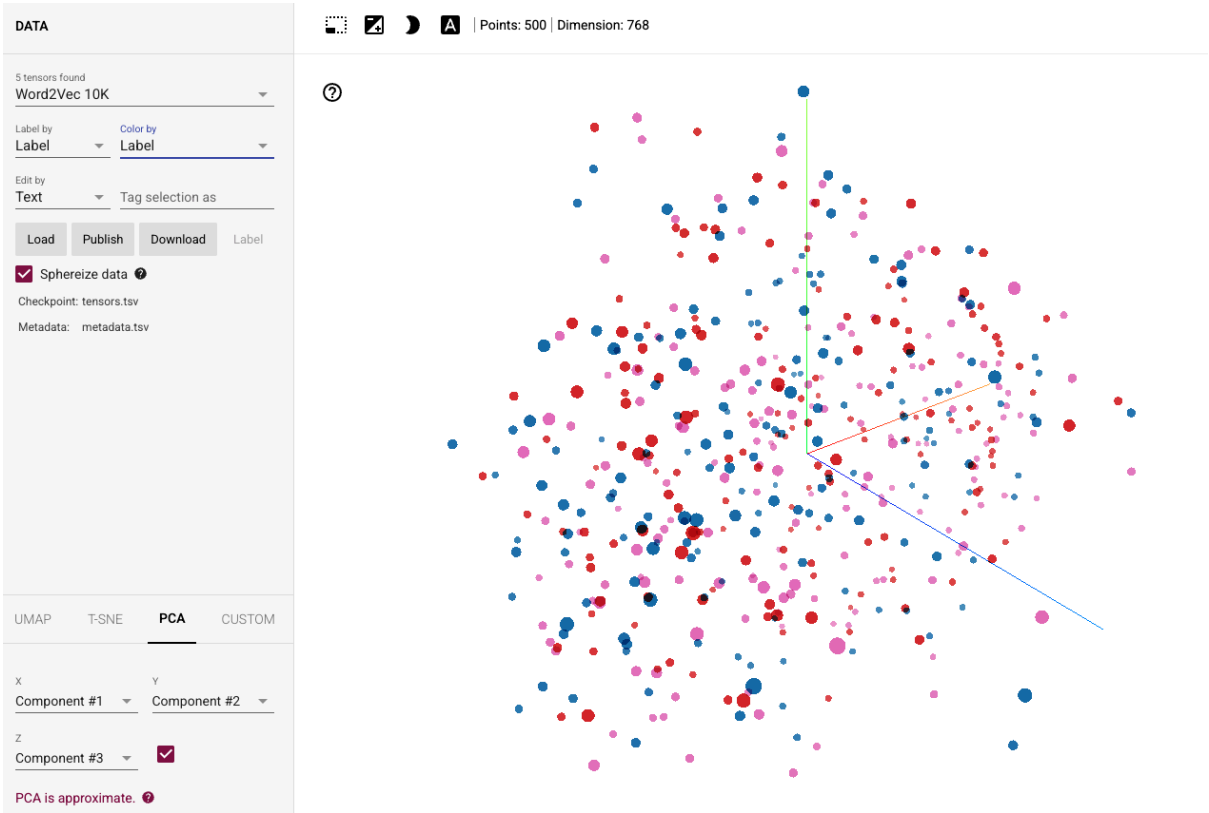

Figure 2: Layer 6 (Intermediate Layer)

At the middle layer, the compact structure dissolves. The vectors spread out, scattering across the representation space. However, the classes (shown in different colors) remain heavily intermixed, illustrating a transition phase in which the model has not yet formed clear decision boundaries.

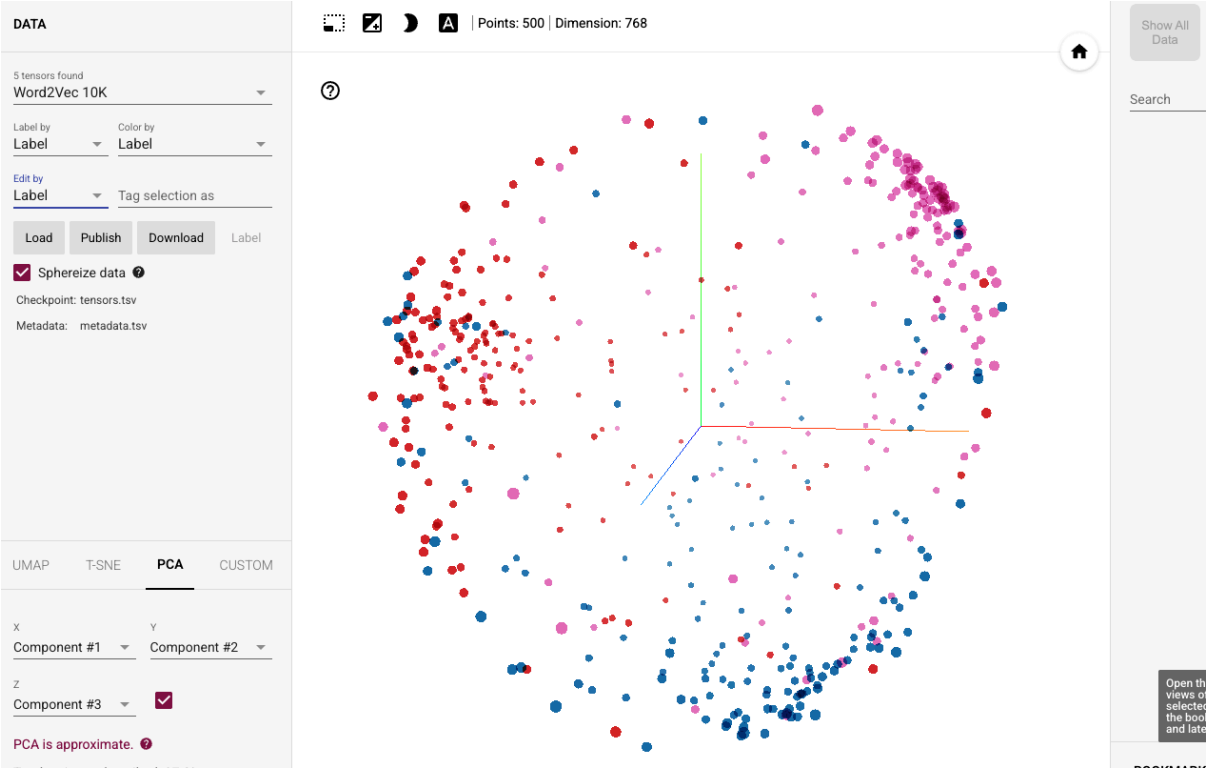

Figure 3: Layer 12 (Output Layer - Final)

In the final output layer, the fine-tuning effect is clearly visible. The scattered mid-layer representations have organized into three distinct clusters corresponding to the NLI labels. While there are overlapping regions at the boundaries (consistent with the ~34% error rate in Turkish), this separation confirms that the model has learned to map complex logical relationships into distinct regions of the representation space, mapping English and Turkish examples into shared semantic clusters.